# CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
## 4th Semester of B. Pharm. Examination
## University Theory Examination October 2017
## PH223 Pharmaceutical Biochemistry

**Date: 4.10.2017, Wednesday** **Time: 1.30 pm to 4.30 pm** **Maximum Marks: 80**

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**Instructions:**

1. **There are three sections in this question paper.**
2. **SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.**
3. **Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.**
4. **Figures to right indicate marks.**
5. **Draw neat labeled sketches wherever necessary.**

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**SECTION - I**

**Q 1** Attempt all questions. Each question is of one mark. **20**

**1.** Mitochondrial DNA is
(A) Circular double stranded
(B) Circular single stranded
(C) Linear double helix
(D) None of these

**2.** The exchange of material takes place
(A) Only by diffusion
(B) Only by active transport
(C) Only by pinocytosis
(D) All of these

**3.** In glucose the orientation of the —H and —OH groups around the carbon atom 5adjacent to the terminal primary alcohol carbon determines
(A) D or L series
(B) Dextro or levorotatory
(C) α and β anomers
(D) Epimers

**4.** A positive Seliwanoff's test is obtained with
(A) Glucose
(B) Fructose
(C) Lactose
(D) Maltose

**5.** A carbohydrate, commonly known as dextrose is
(A) Dextrin
(B) D-Fructose
(C) D-Glucose
(D) Glycogen

**6.** Conversion of glucose to glucose-6-phosphate in human liver is by
(A) Hexokinase only
(B) Glucokinase only
(C) Hexokinase and glucokinase
(D) Glucose-6-phosphate dehydrogenase

**7.** Non essential amino acids
(A) Are not components of tissue proteins
(B) May be synthesized in the body from essential amino acids
(C) Have no role in the metabolism
(D) May be synthesized in the body in diseased state

8. In proteins the $\alpha$-helix and $\beta$-pleated sheet are examples of
(A) Primary structure
(B) Secondary structure
(C) Tertiary structure
(D) Quaternary structure

9. The enzyme carbamoyl phosphate synthetase requires
(A) $Mg^{++}$
(B) $Ca^{++}$
(C) $Na^{+}$
(D) $K^{+}$

10. The chief protein of cow's milk is
(A) Albumin
(B) Vitellin
(C) Livetin
(D) Casein

11. The amino acid which contains an indole group is
(A) Histidine
(B) Arginine
(C) Glycine
(D) Tryptophan

12. The cholesterol molecule is
(A) Benzene derivative
(B) Quinoline derivative
(C) Steroid
(D) Straight chain acid

13. A high fibre diet is associated with reduced incidence of
(A) Cardiovascular disease
(B) C.N.S. disease
(C) Liver disease
(D) Skin disease

14. The major storage form of lipids is
(A) Esterified cholesterol
(B) Glycerophospholipids
(C) Triglycerides
(D) Sphingolipids

15. The fatty acids containing even number and odd number of carbon atoms as well as the unsaturated fatty acids are oxidized by
(A) $\alpha$-oxidation
(B) $\beta$-oxidation
(C) $\omega$-oxidation
(D) All of these

16. Fischer's 'lock and key' model of the enzyme action implies that
(A) The active site is complementary in shape to that of substance only after interaction.
(B) The active site is complementary in shape to that of substance
(C) Substrates change conformation prior to active site interaction
(D) The active site is flexible and adjusts to substrate

17. An example of hydrogen transferring coenzyme is
(A) Co-A
(B) $NAD^{+}$
(C) Biotin
(D) TPP

18. Enzymes which catalyse binding of two substrates by covalent bonds are known as
(A) Lyases
(B) Hydrolases
(C) Ligases
(D) Oxidoreductases

19. DNA does not contain
(A) Thymine
(B) Adenine
(C) Uracil
(D) Deoxyribose

20. In DNA, three hydrogen bonds are formed between
(A) Adenine and guanine
(B) Adenine and thymine
(C) Guanine and cytosine

(D) Thymine and cytosine

## SECTION – II

| | | |
|---|---|---|
| **Q 2** | Attempt any **TWO** of the following | |
| | Define following terms: | |
| **A** | (i) Gluconeogenesis (ii) Transamination (iii) Antagonists (iv) Intrinsic activity (v) Biological oxidation | **05** |
| **B** | Explain chemiosmotic theory of oxidative phosphorylation. | **05** |
| **C** | Describe a pathway for glycolysis. | **05** |
| **Q 3** | Attempt any **TWO** of the following | |
| **A** | Give a comparison of prokaryotic and eukaryotic cells. | **05** |
| **B** | Give a detailed classification of carbohydrates with suitable examples in each class. | **05** |
| **C** | Classify amino acids on the basis of structure and nutritional requirements. | **05** |

## SECTION – III

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|---|---|---|
| **Q 4** | Attempt any **FOUR** of the following | |
| **A** | Add a note on couplers and uncouplers of oxidative phosphorylation. | **05** |
| **B** | Explain metabolic pathway for purine nucleotides. | **05** |
| **C** | Draw a complete pathway for TCA cycle. | **05** |
| **D** | Enumerate various stages of urea cycle and explain its inter relation with Kreb's cycle. | **05** |
| **E** | Add a note on enzyme inhibition. | **05** |
| **F** | Write a note on deamination reactions. | **05** |
| **Q 5** | Attempt any **FOUR** of the following | |
| **A** | Draw a pathway for cholesterol biosynthesis. | **05** |
| **B** | Explain Watson and Crick model of DNA with well labled figure. | **05** |
| **C** | Explain HMP shunt. | **05** |
| **D** | Name various cofactors for oxidation- reduction reactions and explain any one in detail. | **05** |
| **E** | Enumerate various factors affecting rate of enzyme activity and explain any two in detail. | **05** |
| **F** | Add a note on $\beta$- oxidation of fatty acids. | **05** |

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